

PENATONE INJECTION 4MG/ML (2ML AMP)

DESCRIPTION:

A clear, almost colourless sterile solution.

COMPOSITION:

Each ml contains Dexamethasone Sodium Phosphate equivalent to Dexamethasone Phosphate 4mg.

PHARMACODYNAMICS:

Dexamethasone is a glucocorticoid and has the general properties described under corticosteroids; 750 µg of dexamethasone is equivalent in anti-inflammatory activity to about 5 mg of prednisolone.

PHARMACOKINETICS:

Dexamethasone is readily absorbed from the gastro-intestinal tract, its biological half-life in plasma is about 190 minutes. Binding of dexamethasone to plasma proteins is less than for most other corticosteroids. Corticosteroids are metabolised mainly in the liver but also in the kidney, and are excreted in the urine. Dexamethasone, the synthetic glucocorticoid is more slowly metabolised with a lower protein-binding affinity and may account for its increased potency compared with the natural corticosteroid.

INDICATIONS:

By intravenous or intramuscular injection when oral therapy is not feasible: Adrenocortical insufficiency: Dexamethasone phosphate Injection has predominantly glucocorticoid activity with low mineralocorticoid activity. Therefore, it does not offer complete replacement therapy, and its use must be supplemented with salt and/or desoxycorticosterone. When so supplemented, Dexamethasone Phosphate Injection is indicated in the impairment of all adrenocortical activity, as in Addison's disease or following bilateral adrenalectomy that requires replacement of both glucocorticoid and mineralocorticoid activity.

Relative Adrenocortical Insufficiency: In the relative adrenocortical insufficiency that may occur following cessation of long-term therapy with suppressive doses of adrenocortical hormones, mineralocorticoid secretion may be unimpaired. Replacement with a hormone that acts predominantly as a glucocorticoid may be sufficient to restore adrenocortical function. When immediate support is mandatory, Dexamethasone Phosphate Injection may be effective within minutes after administration and can be life-saving.

Pre-operative and Post-operative support: Patients undergoing bilateral adrenalectomy, or hypophysectomy, or any other surgical procedure when adrenocortical reserve is doubtful, and in post-operative shock unresponsive to conventional therapy.

Non-supportive Thyroiditis: Shock: Injection Dexamethasone Phosphate is recommended for the adjunctive therapy of shock where high (Pharmacological) doses of corticosteroids are needed; severe shock of haemorrhagic, traumatic or surgical origin. Treatment with Injection Dexamethasone Phosphate is an adjunct, to and not a substitute for, specific or supportive measures that the patient may require.

Rheumatic Disorders: As adjunctive therapy for short-term administration to support the patient during an acute episode or exacerbation in: Post-traumatic osteoarthritis; synovitis of osteoarthritis; rheumatoid arthritis, including juvenile rheumatoid arthritis (selected cases may require low-dose maintenance therapy); acute subacute bursitis; epicondylitis; acute non-specific tenosynovitis; acute gouty arthritis; psoriatic arthritis; ankylosing arthritis; ankylosing spondylitis.

Collagen disease: During an exacerbation or as maintenance therapy in selected cases of: systemic lupus erythematosus; acute rheumatic carditis.

Dermatological Disease: Pemphigus; severe erythema multiforme (Stevens Johnson Syndrome); exfoliative dermatitis; bullous dermatitis herpetiformis; severe seborrheic dermatitis; severe psoriasis; mycosis fungoides.

Allergic states: Control of severe of incapacitating allergic conditions intractable to adequate trials of conventional treatment in: bronchial asthma; contact dermatitis; atopic dermatitis; serum sickness; seasonal or perennial allergic rhinitis; drug hypersensitivity reactions; acute noninfectious laryngeal oedema; anaphylaxis (epinephrine is the drug of first choice).

Ophthalmic Diseases: Severe acute and chronic allergic and inflammatory processes involving the eye and its adnexa such as: allergic conjunctivitis; keratitis; allergic corneal marginal ulcers; herpes zoster ophthalmicus; iritis, iridocyclitis; chorioretinitis; diffuse posterior uveitis and choroiditis; optic neuritis; sympathetic ophthalmia; anterior segment inflammation.

Gastrointestinal diseases: To support the patient during a critical period of disease in: ulcerative colitis (systemic therapy); regional enteritis (systemic therapy); regional enteritis (systemic therapy).

Respiratory Disease: Symptomatic sarcoidosis: Loeffler's syndrome not manageable by other means; Berylliosis; fulminating or disseminated pulmonary tuberculosis when concurrently accompanied by appropriate antituberculous chemotherapy; aspiration pneumonia.

Haematological Disorders: Acquired (autoimmune) haemolytic anaemia; idiopathic thrombocytopenic purpura in adults (I.V. only, I.M. administration is contraindicated); secondary thrombocytopenia in adults; erythroblastopenia (R.B.C. anaemia); congenital (erythroid) hypoplastic anaemia.

Neoplastic Diseases: For palliative management of: hypercalcaemia associated with cancer; leukemias and lymphomas in adults; acute leukemia of childhood.

Oedematous States: To induce diuresis or remission of proteinuria in the nephrotic syndrome without uraemia of the idiopathic type, or that due to lupus erythematosus.

Cerebral oedema: Dexamethasone Phosphate Injection may be used to treat patients with cerebral oedema from various causes: (a) associated with primary or metastatic brain tumours, (b) associated with neurosurgery, (c) associated with head injury or pseudo-tumour cerebri, (d) associated with cerebral vascular accident (acute stroke) involving the cerebral cortex.

It may be used also in the preoperative preparation of patients with increased intracranial pressure secondary to brain tumours or for palliation of patients with inoperable or recurrent brain neoplasms. Use of Dexamethasone Phosphate Injection in cerebral oedema is not a substitute for careful neurological evaluation and definitive management such as neurosurgery or other specific therapy.

Diagnostic Testing of Adrenocortical Hyperfunction

Neonatal respiratory distress syndrome; antenatal prophylaxis: Use of Dexamethasone Phosphate Injection in mothers at high risk for premature delivery has been shown to reduce the incidence of neonatal respiratory distress syndrome.

By intra-articular or soft-tissue injection: As adjunctive therapy for short-term administration (to support patient during an acute episode or exacerbation) in: synovitis of osteoarthritis; rheumatoid arthritis; acute and subacute bursitis; acute gouty arthritis; epicondylitis; acute non-specific tenosynovitis; post-traumatic osteoarthritis.

By intralesional injection: Keloids: localised hypertrophic, infiltrated, inflammatory lesions of: lichen planus, psoriatic plaques, granuloma annulare, and lichen simplex chronicus (neuro-dermatitis); discoid lupus erythematosus; necrobiosis lipidica diabetorum; alopecia areata; may also be useful in cystic tumours of an aponeurosis or tendon (ganglia).

RECOMMENDED DOSAGE:

This preparation can be given without mixing or dilution. If preferred, it can be added to Sodium Chloride Injection, or Dextrose Injection, or compatible blood for transfusion, without loss of potency, and administered by intravenous drip. Solutions used for intravenous administration or further dilution of this product should be preservative free when used in the neonate, especially in the premature infant. When Dexamethasone Phosphate Injection is added to an infusion solution, the mixture must be used within 24 hours at 25°C since infusion solutions do not contain preservatives. The usual aseptic techniques governing injections should be observed.

Intravenous and Intramuscular Injection: The usual initial dosage of Dexamethasone Phosphate Injection may vary from 0.5 mg to 20 mg per day depending on the specific disease entity being treated. Usually, the parenteral dosage ranges are one-third to one-half the oral dose given every 12 hours. However, in certain overwhelming, acute life-threatening situations, administration in dosages exceeding the usual recommended dosages have been used. In these circumstances, the slower rate of absorption by intramuscular administration should be recognised.

DOSAGE REQUIREMENTS ARE VARIABLE AND MUST BE INDIVIDUALISED ON THE BASIS OF THE DISEASE UNDER TREATMENT AND THE RESPONSE OF THE PATIENT. If the drug is to be stopped after it has been given for more than a few days, it is recommended that it be withdrawn gradually rather than stopped abruptly. In emergencies, the usual dose is 1 mL to 5 mL (4 mg to 20 mg) I.V. or I.M. (in shock use only the I.V. route). This dose may be repeated until adequate response is noted. After initial improvement, single doses of 0.5 mL to 1 mL (2 mg to 4 mg), should be repeated as necessary, may be sufficient. The total daily dosage usually need not exceed 20 mL (80 mg), even in severe conditions. When constant maximal effect is desired, dosage must be repeated at three-hour or four-hour intervals, or maintained by slow intravenous drip. Intravenous and intramuscular injections are advised in acute illness. When the acute stage has passed, substitute oral steroid therapy as soon as feasible.

Shock (of haemorrhagic, traumatic, surgical origin): The usual dose is 2 to 6 mg/kg body weight given as a single intravenous injection. This may be repeated in 2 to 6 hours, if shock persists. As an alternative Injection Dexamethasone Phosphate, 2 to 6 mg/kg body weight, is given as a single intravenous injection followed immediately by the same dose in an intravenous infusion. Therapy with injection Dexamethasone Phosphate is an adjunct to and not a replacement for conventional therapy (see Precautions). Administration of high dose corticosteroid therapy should be continued only until the patients condition has stabilised and usually no longer than 48 to 72 hours.

Cerebral oedema associated with primary or metastatic brain tumour, neurosurgery, head injury, pseudotumour cerebri or preoperative preparation of patients with increased intracranial pressure secondary to brain tumour: Initially 10 mg (2.5 mL) Dexamethasone Phosphate Injection intravenously followed by 4 mg (1 mL) intramuscularly every 6 hours until symptoms of cerebral oedema subside. Response is usually noted within 12 to 24 hours; dosage may be reduced after 2 to 4 days and gradually discontinued over a period of 5 to 7 days. High doses of Injection Dexamethasone Phosphate are recommended for initiating short-term intensive therapy for acute life-threatening cerebral oedema. Following the high loading dose schedule of the first day of therapy, the dose is scaled down over the 7-10 day period of intensive therapy and subsequently reduced to zero over the next 7-10 days. When maintenance therapy is required, this should be changed to oral corticosteroid therapy as soon as possible.

Suggested high dose schedule in cerebral oedema (See chart below):

Suggested high dose schedule in cerebral oedema	
Adults	
Initial Dose	50 mg I.V.
1 st day	8 mg I.V. every 2 hours
2 nd day	8 mg I.V. every 2 hours
3 rd day	8 mg I.V. every 2 hours
4 th day	4 mg I.V. every 2 hours
5 th -8 th day	4 mg I.V. every 4 hours
Thereafter	Decrease by daily reduction of 4 mg

Children (35 kg and over)	
Initial dose	25 mg I.V.
1 st day	4 mg I.V. every 2 hours
2 nd day	4 mg I.V. every 2 hours
3 rd day	4 mg I.V. every 2 hours
4 th day	4 mg I.V. every 4 hours
5 th -8 th day	4 mg I.V. every 6 hours
Thereafter	Decrease by daily reduction of 2 mg
Children (below 35 kg)	
Initial dose	20 mg I.V.
1 st day	4 mg I.V. every 3 hours
2 nd day	4 mg I.V. every 3 hours
3 rd day	4 mg I.V. every 3 hours
4 th day	4 mg I.V. every 6 hours
5 th -8 th day	2 mg I.V. every 6 hours
Thereafter	Decrease by daily reduction of 1 mg

For palliative management of patients with recurrent or inoperable brain tumours: Maintenance therapy should be individualised with Dexamethasone Phosphate Injection, or oral corticosteroid therapy. A dosage of 2 mg, 2 to 3 times a day may be effective.

Associated with acute stroke (excluding intracerebral haemorrhage): Initially 10 mg (2.5 mL) Dexamethasone Phosphate Injection intravenously followed by 4 mg (1 mL) intramuscularly every 6 hours for 10 days. Dose should then be tapered to zero over the ensuing 7 days. The smallest dosage necessary to control cerebral oedema should be utilised.

Intra-articular, intralesional and soft-tissue injection: Intra-articular, intralesional, and soft-tissue injections generally are employed when affected joints or areas are limited to one or two sites.

Some of the usual single doses are:

Site of Injection	Volume of Injection (ml)	Amount of Dexamethasone Phosphate (mg)
Large joints (eg. knee)	0.5 to 1	2 to 4
Small joints (eg. interphalangeal, temporomandibular)	0.2 to 0.25	0.8 to 1
Bursae	0.5 to 0.75	2 to 3
Tendon sheaths	0.1 to 0.25	0.4 to 1
Soft tissue infiltration	0.5 to 1.5	2 to 6
Ganglia	0.25 to 0.5	1 to 2

The frequency of injection varies from once every 3 to 5 days to once every 2 to 3 weeks, depending on the response to treatment.

Neonatal respiratory distress syndrome: antenatal prophylaxis: The recommended dosage of Dexamethasone Phosphate Injection is 5 mg (1.25 mL) administered intramuscularly to the mother every twelve hours for up to a total of four doses. Administration should be initiated preferably between 24 hours and seven days before the estimated delivery.

ROUTE OF ADMINISTRATION:

For i.v., i.m., intra-synovial and soft tissue injection.

CONTRAINDICATIONS:

Systemic fungal infections (see **Precautions** Amphotericin B). Hypersensitivity to sulphites of any other component of this medication (see **Precautions**). Administration of live virus vaccines (see **Precautions**).

WARNING & PRECAUTIONS:

This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites. The overall prevalence of sulphite sensitivity in the general population is unknown and probably low. Sulphite sensitivity is seen more frequently in asthmatic than in non-asthmatic people. Corticosteroids may exacerbate systemic fungal infections and therefore should not be used in the presence of such infections unless they are needed to control drug reactions due to amphotericin B. Moreover, there have been cases reported in which concomitant use of amphotericin B and hydrocortisone was followed by cardiac enlargement and congestive failure.

Literature reports suggest an apparent association between use of corticosteroids and left ventricular free wall rupture after a recent myocardial infarction, therefore therapy with corticosteroids should be used with great caution in these patients. Average and large doses of cortisone or hydrocortisone can cause elevation of blood pressure, salt and water retention, and increased excretion of potassium. These effects are less likely to occur with the synthetic derivatives except when used in large doses. Dietary salt restriction and potassium supplementation may be necessary. All corticosteroids increase calcium excretion, drug-induced secondary adrenocortical insufficiency may result from too rapid withdrawal of corticosteroids and may be minimised by gradual reduction of dosage.

This type of relative insufficiency may persist for months after discontinuation of therapy; therefore, in any situation of stress occurring during that period, corticosteroid therapy should be reinstituted or the current dosage may have to be increased. Since mineralocorticoid secretion may be impaired, salt and/or a mineralocorticoid should be administered concurrently. Following prolonged therapy, withdrawal of corticosteroids may result in symptoms of the corticosteroid withdrawal syndrome, including fever, myalgia, arthralgia and malaise. This may occur in patients even without evidence of adrenal insufficiency. Because rare instances of anaphylactoid reactions have occurred in patients receiving parenteral corticosteroid therapy, appropriate precautionary measures should be taken prior to administration, especially when the patient has a history of allergy to any drug. Administration of live virus vaccines, including smallpox, is contraindicated in individuals receiving immunosuppressive doses of corticosteroids. If inactivated viral or bacterial vaccines are administered to individuals receiving immunosuppressive doses of corticosteroids, the expected serum antibody response may not be obtained. However, immunisation procedures may be undertaken in patients who are receiving corticosteroids as replacement therapy, eg. for Addison's disease. The use of Dexamethasone Phosphate Injection in active tuberculosis should be restricted to those cases of fulminating or disseminated tuberculosis in which the corticosteroid is used for the management of the disease in conjunction with an appropriate antituberculous regimen. If corticosteroid are indicated in patients with latent tuberculosis or tuberculin reactivity, close observation is necessary as reactivation of the disease may occur. During prolonged corticosteroid therapy, these patients should receive chemoprophylaxis.

Steroids should be used with caution in non-specific ulcerative colitis, if there is a probability of impending perforation, abscess, or other pyrogenic infection, diverticulitis, fresh intestinal anastomoses, active or latent peptic ulcer, renal insufficiency, hypertension, osteoporosis, and myasthenia gravis, signs of peritoneal irritation following gastrointestinal perforation in patients receiving large doses of corticosteroids may be minimal or absent. Fat embolism has been reported as a possible complication of hypercorticism. Corticosteroids should be used cautiously in patients with ocular herpes simplex because of possible corneal perforation. There is an enhanced effect of corticosteroids in patients with hypothyroidism and in those with cirrhosis. Steroids may increase or decrease motility and number of spermatozoa in some patients. Corticosteroids may mask some signs of infection, and new infections may appear during their use. In cerebral malaria the use of corticosteroids is associated with prolongation of coma and a higher incidence of pneumonia and gastrointestinal bleeding. Corticosteroids may activate latent amoebiasis. Therefore, it is recommended that latent or active amoebiasis be ruled out before initiating corticosteroid therapy in any patient who has spent time in the tropics or in any patient with unexplained diarrhoea.

Prolonged use of corticosteroids may produce posterior subcapsular cataracts glaucoma with possible damage to the optic nerves, and may enhance the establishment of secondary ocular infections due to fungi or viruses. Growth and development of infants and children on prolonged corticosteroid therapy should be carefully observed. Intra-articular injection or a corticosteroid may produce systemic as well as local effects. A marked increase in pain accompanied by local swelling, further restriction of joint motion, fever and malaise are suggestive of septic arthritis. If this complication occurs and the diagnosis of sepsis is confirmed, appropriate antimicrobial therapy should be instituted. Local injection of a steroid into an infected site is to be avoided. Appropriate examination of any joint fluid present is necessary to exclude a septic process. Corticosteroids should not be injected into unstable joints. Frequent intra-articular injection may result in damage to joint tissues. Patients should be impressed strongly with the importance of not overusing joints in which symptomatic benefit has been obtained as long as the inflammatory process remains active.

Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk/benefit evaluation.

INTERACTION WITH OTHER MEDICATIONS:

Acetylsalicylic acid should be used cautiously in conjunction with corticosteroids in hypoprothrombinaemia, diphenylhydantoin (phenytoin), phenobarbital, ephedrine, and rifampicin may enhance the metabolic clearance of corticosteroids, resulting in decreased blood levels and lessened physiologic activity, thus requiring adjustment in corticosteroid dosage. These interactions may interfere with dexamethasone suppression tests, which should be interpreted with caution during administration of these drugs. False-negative results in the dexamethasone suppression test in patients being treated with indomethacin have been reported. The prothrombin time should be checked frequently in patients who are receiving corticosteroids and coumarin anticoagulants at the same time because of reports that corticosteroids have altered the response to these anticoagulants. Studies have shown that the usual effect produced by adding corticosteroids is inhibition of response to coumarins, although there have been some conflicting reports of potentiation not substantiated by studies. When corticosteroids are administered concomitantly with potassium-depleting diuretics, patients should be observed closely for development of hypokalaemia.

Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of system side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects.

USE IN PREGNANCY & LACTATION:

Use in pregnancy: In animal experiments, corticosteroids have been found to cause malformations of various kinds (cleft palate, skeletal malformations). These findings do not seem to be relevant to humans. Reduce placental and birth weight have been recorded in animals and humans after long-term treatment. Since the possibility of suppression of the adrenal cortex in the newborn baby after long-term treatment must be considered, the needs of the mother must be carefully weighed against the risk to the fetus when prescribing corticosteroids. However, the short term use of antepartum

corticosteroids for the prevention of respiratory distress syndrome, when warranted does not pose a risk. Since human reproduction studies have not been done with corticosteroids, the use of these drugs in pregnancy or in women of childbearing potential, requires that the anticipated benefits be weighed against the possible hazards to the mother and embryo or foetus. Infants born to mothers who have received substantial doses of corticosteroids during pregnancy should be carefully observed for signs of hypoadrenalism.

Use in lactation: Corticosteroids appear in breast milk and could suppress growth, interfere with endogenous corticosteroid production, or cause unwanted effects. Mothers taking pharmacological doses of corticosteroids should be advised not to nurse. Corticosteroids may affect the nitroblue tetrazolium test for bacterial infection and produce false negative results.

SIDE EFFECTS:

Fluid and electrolyte disturbances: Sodium retention, fluid retention, congestive heart failure in susceptible patients, potassium loss, hypokalaemic alkalosis, hypertension.

Musculoskeletal: Muscle weakness, steroid myopathy, loss of muscle mass, osteoporosis, vertebral compression fractures, aseptic necrosis of femoral and humeral heads, pathological fracture of long bones, tendon rupture.

Gastrointestinal: Peptic ulcer with possible perforation and haemorrhage, perforation of the small and large bowel, particularly in patients with inflammatory bowel disease, pancreatitis, abdominal distention, ulcerative oesophagitis.

Dermatological: Impaired wound healing: Thin fragile skin, petechiae and ecchymoses, erythema, increased sweating, may suppress reactions to skin tests, burning or tingling, especially in the perineal area, after I.V. injection, other cutaneous reactions, such as allergic dermatitis, urticaria, angioneurotic oedema.

Neurological: Convulsions, increased intracranial pressure with papilloedema (pseudotumour cerebri) usually after treatment, vertigo, headache, psychic disturbances.

Endocrine: Menstrual irregularities, development of cushingoid state, suppression of growth in children, secondary adrenocortical and pituitary unresponsiveness, particularly in times of stress, as in trauma, surgery or illness, decreased carbohydrate tolerance, manifestations of latent diabetes mellitus, increased requirements for insulin or oral hypoglycaemic agents in diabetes, hirsutism.

Ophthalmic: Posterior subcapsular cataracts, increased intraocular pressure, glaucoma, exophthalmos.

Metabolic: Negative nitrogen balance due to protein catabolism.

Cardiovascular: Myocardial rupture following recent myocardial infarction (see Precautions).

Other: Anaphylactoid or hypersensitivity reactions, thromboembolism, weight gain, increased appetite, nausea, malaise, hiccups.

The following additional adverse reactions are related to parenteral corticosteroid therapy: Rare instances of blindness associated with intralesional therapy around the face and head; hyperpigmentation or hypo-pigmentation; subcutaneous and cutaneous atrophy; sterile abscess; post-infection flare (following intra-articular use); Charcot-like arthropathy.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Treatment of adverse effects of corticosteroids: The adverse effects of corticosteroids are nearly always due to their use in excess of normal physiological requirements. They should be treated symptomatically, where possible the dosage being reduced or the drug slowly withdrawn. Acute adrenal insufficiency in corticosteroid-treated patients, whether it be due to accidental abrupt withdrawal of the corticosteroid or the inability of the patient's adrenals to cope with the increased stress of infection or accidental or surgical trauma, must be treated immediately with intravenously injections of hydrocortisone sodium succinate together with infusions of dextrose in 0.9% sodium chloride. Hydrocortisone sodium succinate 100 to 300 mg should be given intravenously immediately followed by 100 mg intravenously every eight hours. Infusions of dextrose in 0.9% sodium chloride solution should be given in amounts adequate to correct hypoglycaemia, sodium loss, and dehydration; special care must be taken, however, not to overload the circulation. Suitable antibiotic therapy is indicated where the underlying cause is an infection. As the patient recovers, the dose of hydrocortisone sodium succinate may be changed to the intramuscular route before gradually tapering off and re-establishing the patient's original corticosteroid therapy.

INCOMPATIBILITIES:

Dexamethasone sodium phosphate is physically incompatible with daunorubicin, doxorubicin and vancomycin and should not be admixed with solution containing these drugs. Also incompatible with doxapram HCl and glycopyrrolate in syringe.

STORAGE CONDITIONS:

Store below 30° C. Protect from freezing and light. KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Packs of 10x2 ml ampoules

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.